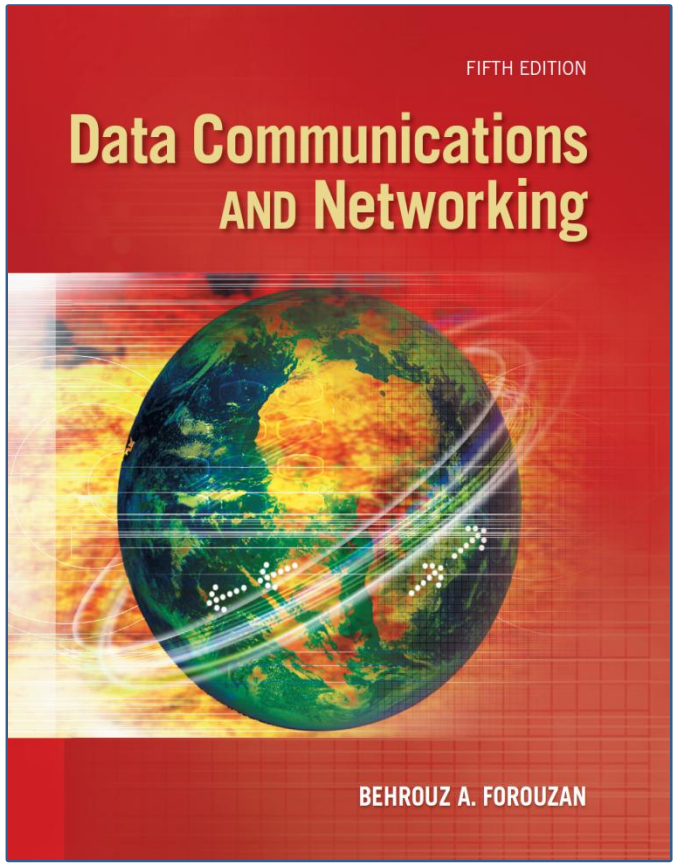


Network Services

Lecturer: Dr. Khaled BinAbdl

Lecture 2

Domain Name System

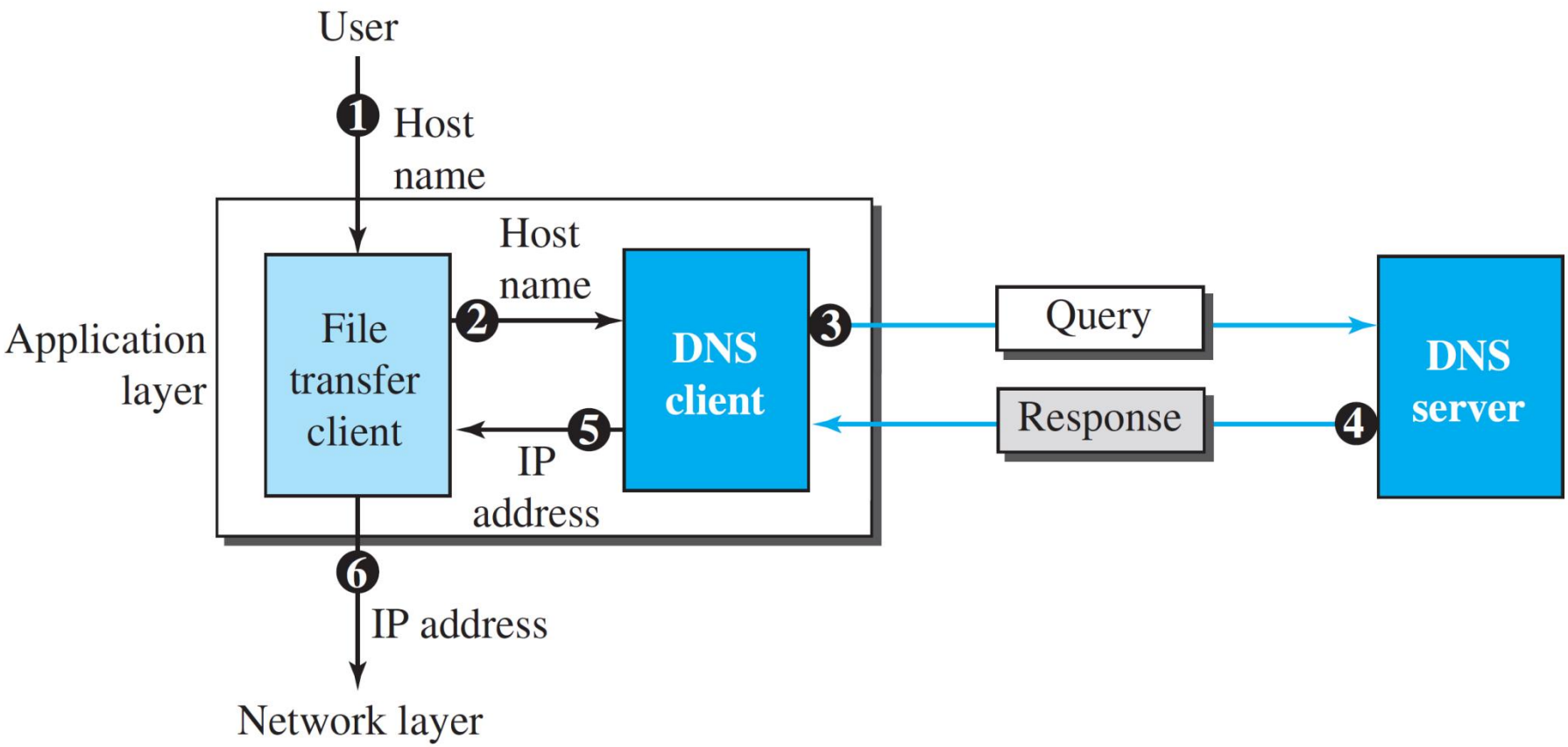
- TCP/IP **protocols** use the **IP** address, which uniquely **identifies** the connection of a host to the Internet.
- **People** prefer to use **names** instead of numeric addresses.
- A **directory** system in the Internet can map **names** to **IP** addresses like **telephone** directory.
- Internet is so **huge** today, a **central** directory system **cannot hold** all the mapping.

Domain Name System cont.

- If the **central** computer **fails**, the whole communication network will **collapse**.
- A **better** solution is to **distribute** the information among **many** computers in the world.
- the **host** that needs mapping can **contact** the **closest** computer holding the needed information.
- This method is used by the **Domain Name System (DNS)**.

The Mapping Process in Domain Name System

Purpose of DNS



The Mapping Steps

The following 6 steps map the host name to an IP address:

1. The user passes the host **name** to the **file transfer client**.
2. The **file transfer client** passes the host name to the **DNS** client.
3. Each computer, after being booted, knows the address of **one DNS server**. The DNS client sends a message to a DNS server with a query that gives the file transfer server name using the known IP address of the DNS server.
4. The DNS server **responds** with the **IP address** of the desired file transfer server.
5. The DNS server passes the IP address to the file transfer client.
6. The file transfer client now uses the received IP address to access the file transfer server.

Name Space

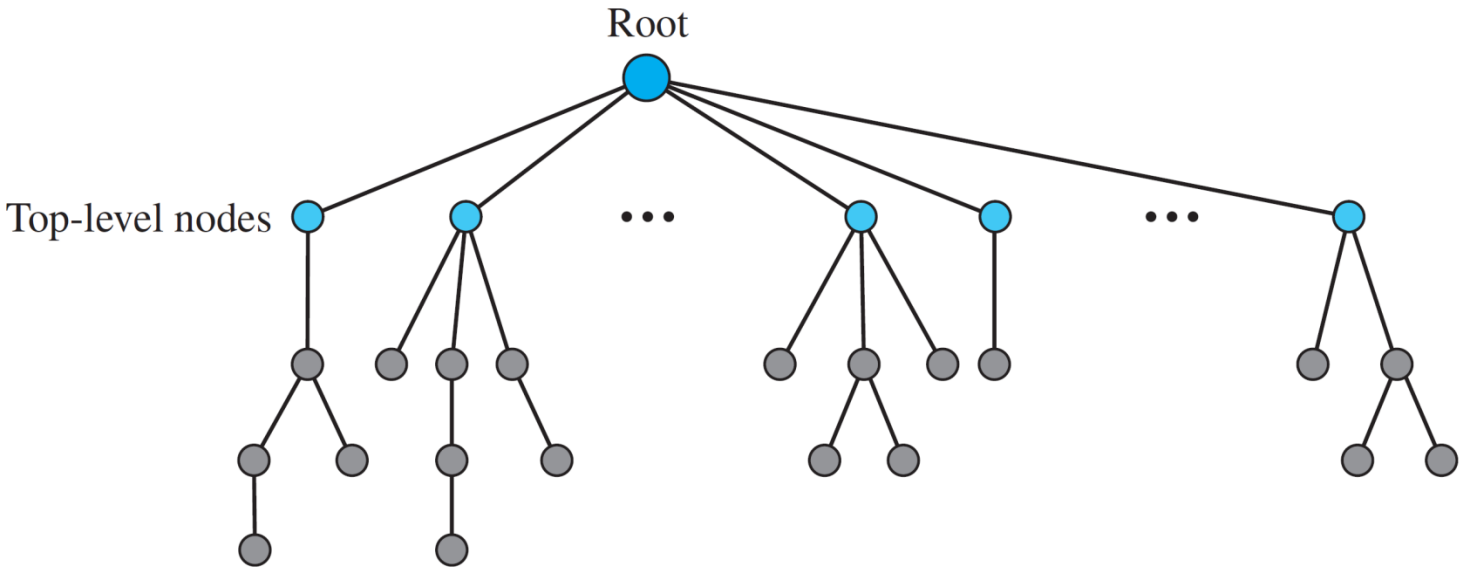
– A **name space** that maps each address to a unique name can be organized in two ways:

- ***flat name space:***
  - A name is **assigned** to an address as a sequence of characters **without structure**.
  - The main **disadvantage** of a flat name space is that it **cannot** be used in a **large** system such as the Internet
- ***hierarchical name space:***
  - Each name is made of **several parts**.
  - The **first part** can define the **nature** of the organization, the **second part** can define the **name** of an organization, the **third part** can define **departments** in the organization.

Domain Name Space

In **Domain Name Space** design the names are defined in an **inverted-tree** structure with the **root** at the **top (**max **128** Levels – 0 - 127**)**.

Domain name space



Domain Name Space cont.

Contents:

1- **Label**:

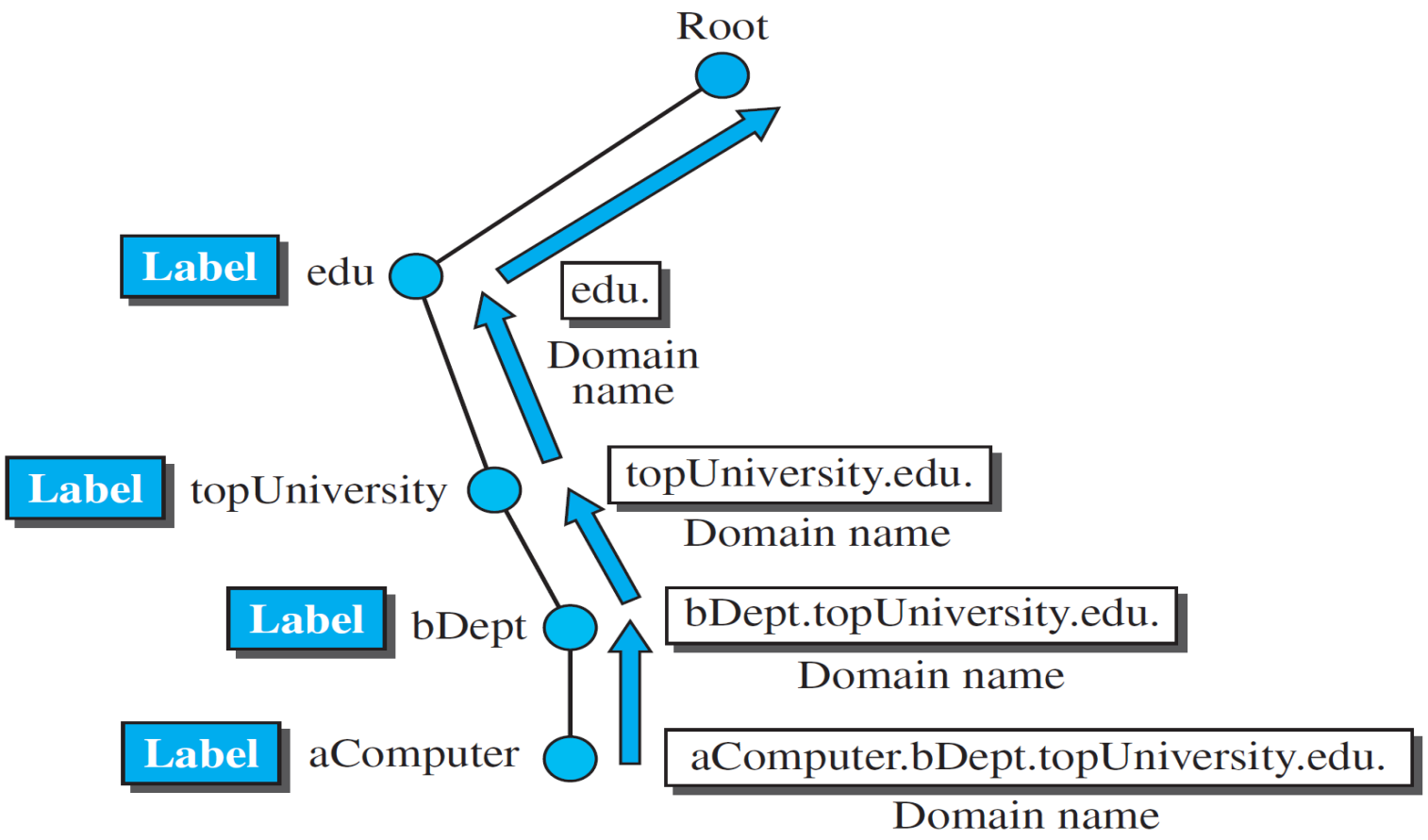
- Each node in the tree has a **label.**
- **Maximum** of **63** characters.
- The **root** label is a **null** string.
- **Children** of a node have **different** label

2- **Domain Name**:

- Each node in the tree has a domain name
- **Domain name** is a sequence of labels separated by dots (**.**)

Domain Name Space cont.

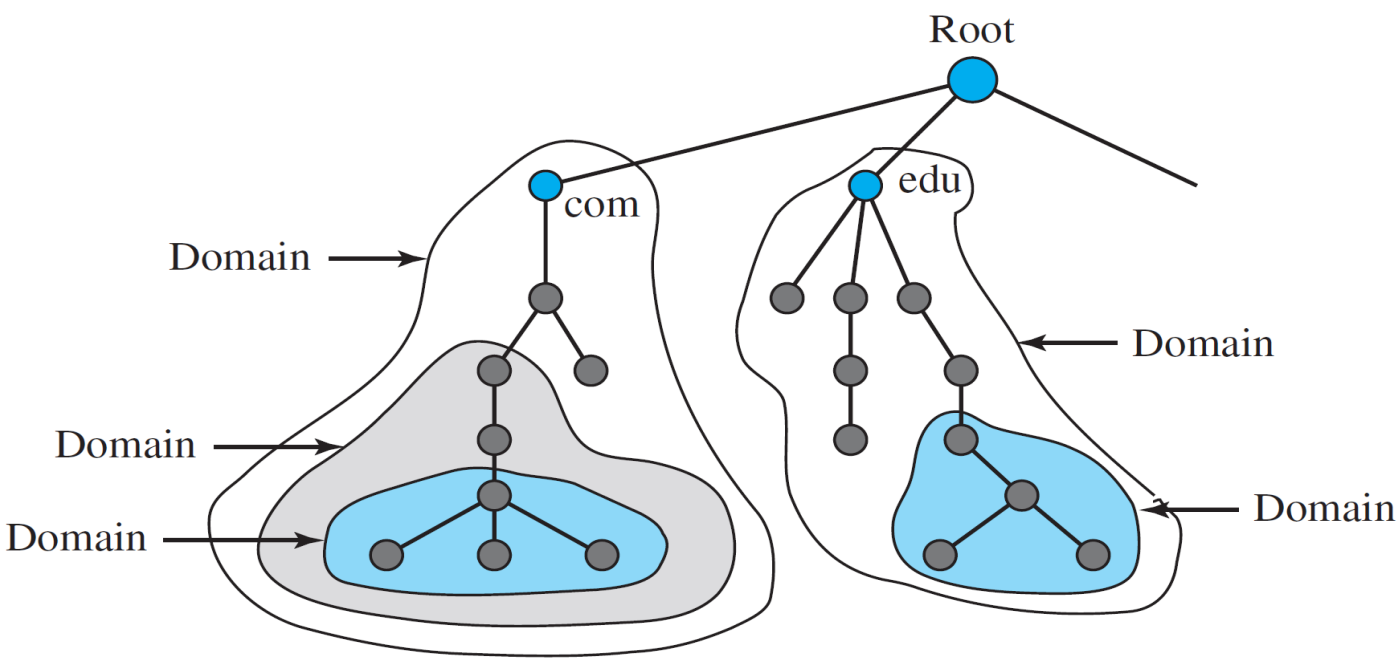
Domain names and labels



Domains

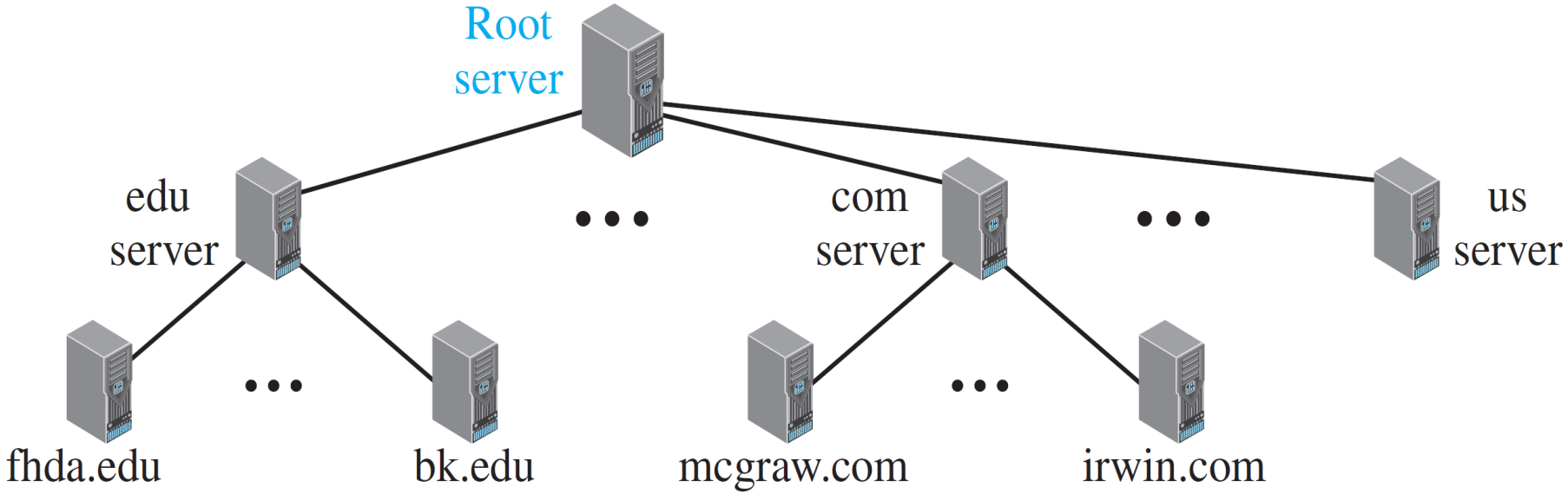
Domain: A **domain** is a subtree of the domain name space.

Domains



Hierarchy of Name Servers

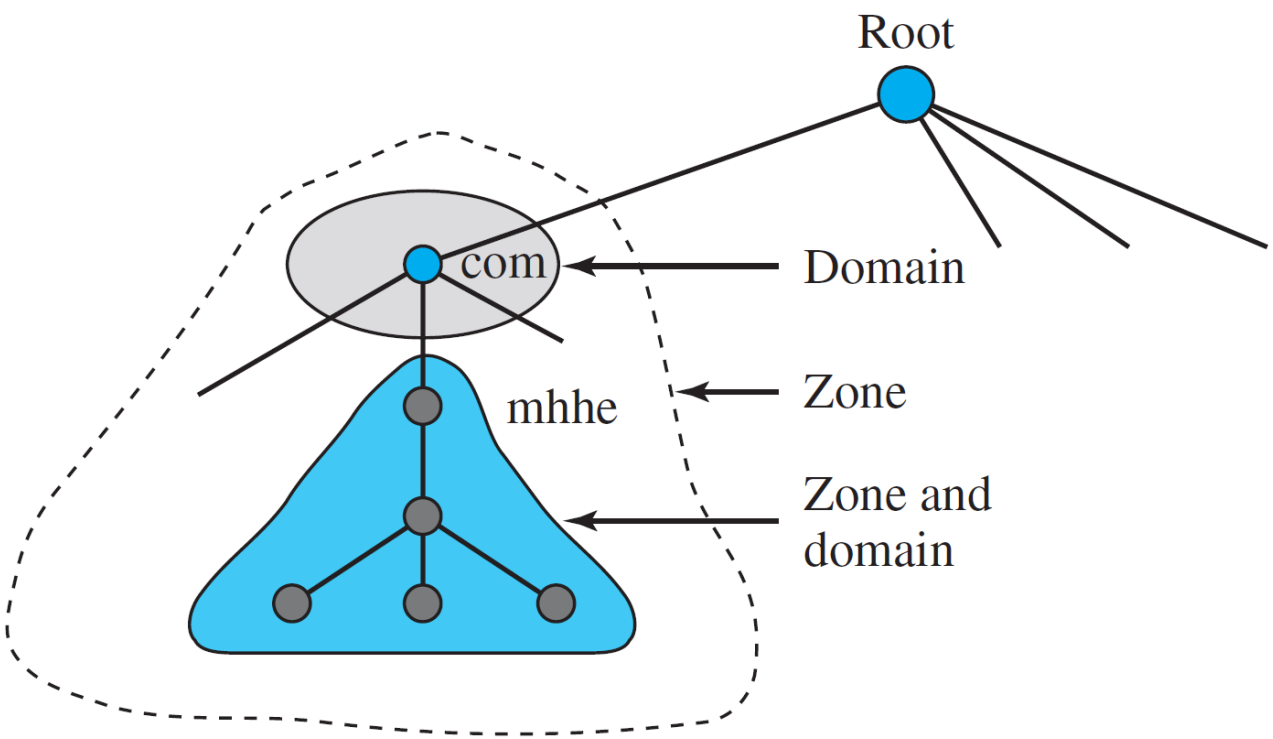
Hierarchy of name servers



Zone

A zone is a contiguous part of the entire tree.

Zone



Root Server

- A **root server** is a server whose zone consists of the whole tree

- A **root** server usually **does not store** any information about domains but **delegates** its authority to **other** servers, keeping **references** to those servers.

Primary and Secondary Servers

- A ***primary server*** is a server that stores a **file** about the **zone** for which it is an **authority**.

- A ***secondary server*** is a server that transfers the complete information about a zone from another server (primary or secondary)

DNS in the Internet

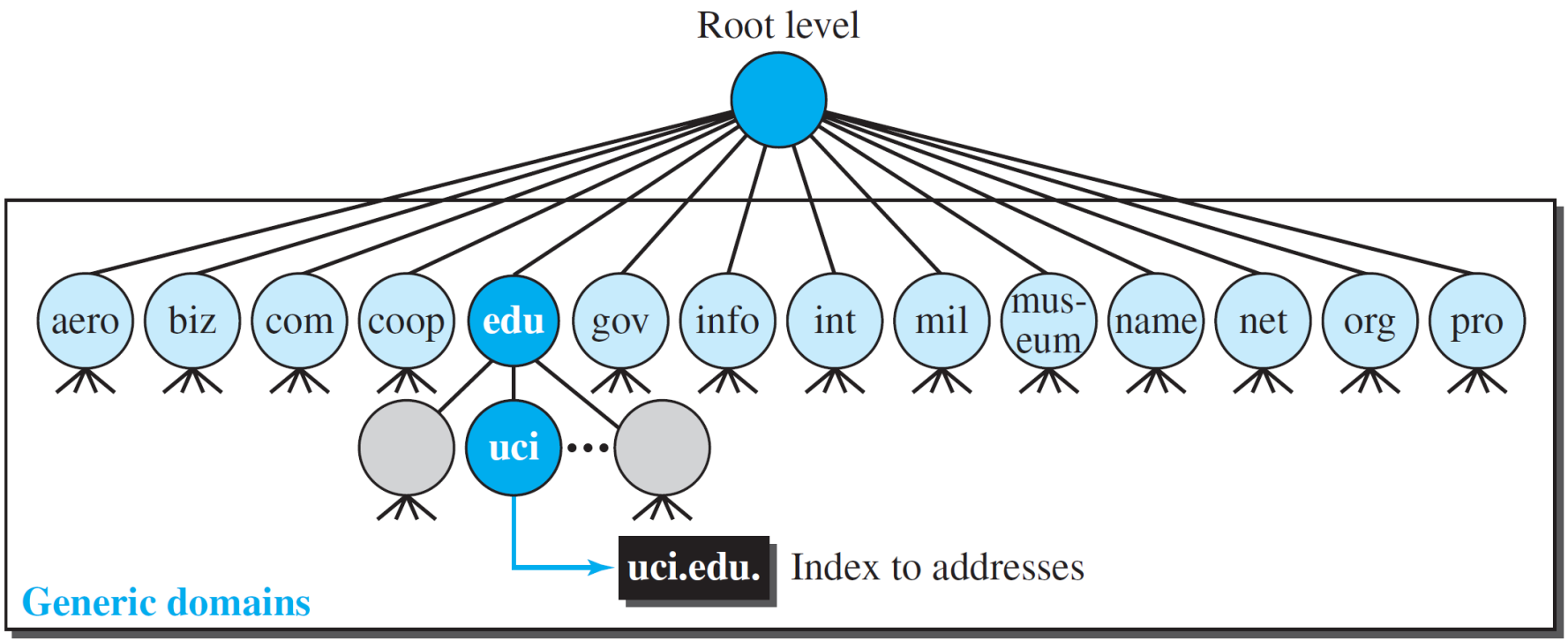
Generic Domains

– The **generic domains** define registered hosts according to their generic behaviour.

– Each node in the tree defines a domain, which is an **index** to the domain name space database.

Generic Domains

Generic domains



Generic Domains Labels

Generic domain labels

Label	*Description*	*Label*	*Description*
aero	Airlines and aerospace	**int**	International organizations
biz	Businesses or firms	**mil**	Military groups
com	Commercial organizations	**museum**	Museums
coop	Cooperative organizations	**name**	Personal names (individuals)
edu	Educational institutions	**net**	Network support centers
gov	Government institutions	**org**	Nonprofit organizations
info	Information service providers	**pro**	Professional organizations

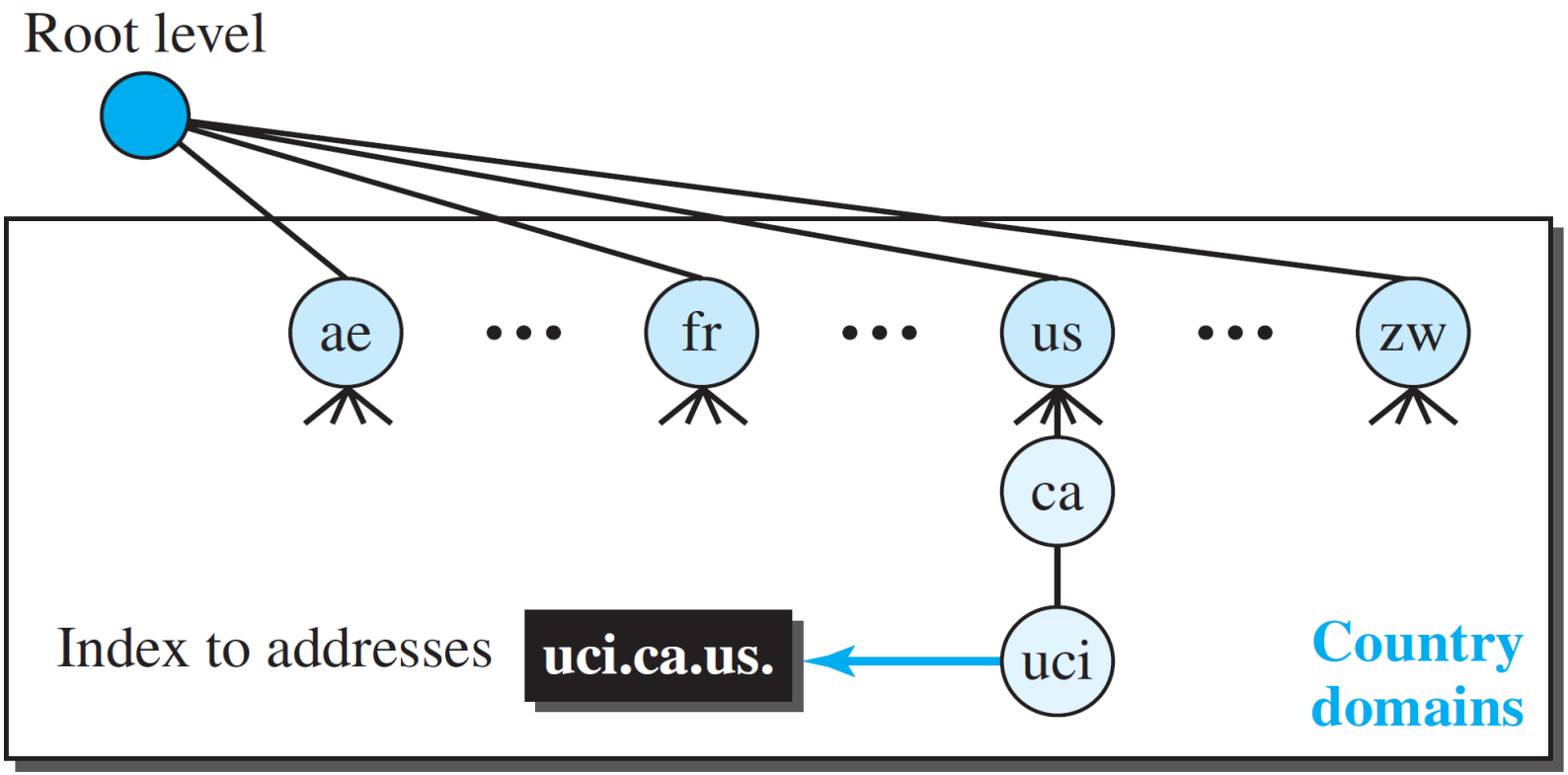
Country Domains

- The **country domains** section uses two-character country abbreviations.

- For Example: The address ***uci.ca.us.*** can be translated to University of California, Irvine, in the state of California in the United States.

Country Domains cont.

Country domains



Resolution

- **Mapping** a name to an address is called ***name-address resolution***.
- A **host** that needs to map an **address** to a name or a name to an address calls a DNS **client** called a ***resolver.***
- The resolver accesses the **closest** DNS server with a mapping request.
- If the server **has** the information, it **satisfies** the resolver;
- otherwise, it either **refers** the resolver to other servers or asks other servers to provide the information.

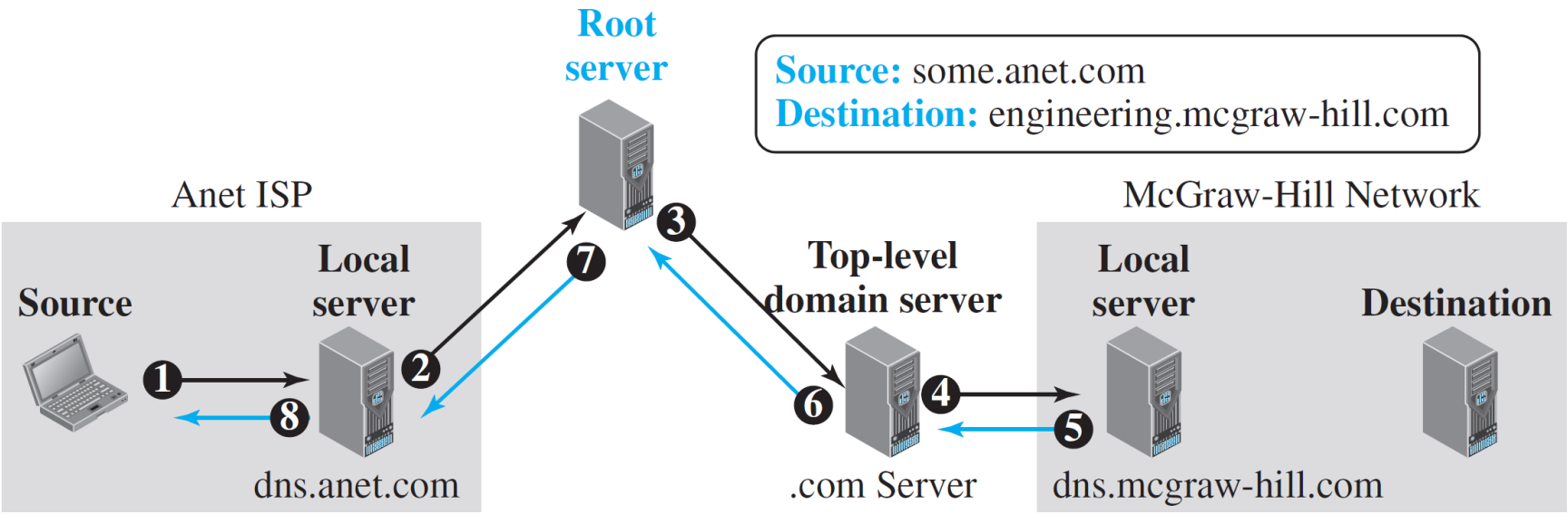
Resolution Types

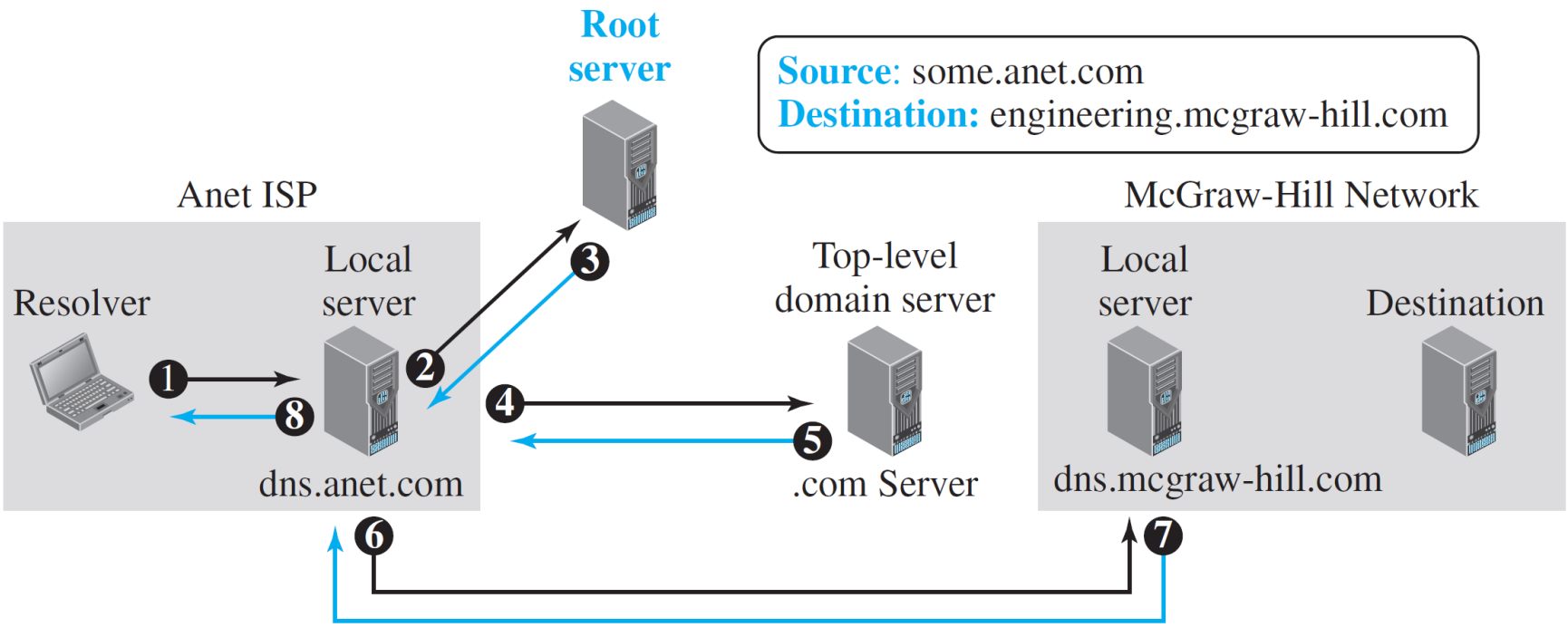
– A resolution can be either:

– **Recursive**
– **Iterative.**

Recursive Resolution

Recursive resolution



Iterative Resolution

Iterative resolution